Core practical – Identifying ions, Part 1

Pearson Edexcel GCSE (9-1) Chemistry

Name _____ **Class** _____ **Date** _____

Part 1: Flame tests

Practical Objective

You will determine the flame test colours for lithium, sodium, potassium and calcium ions. You will then use flame tests to identify the metal ions in different unknown solids.

Apparatus

- eye protection
- Bunsen burner
- heat-resistant mat
- flame test loop
- beaker
- dilute hydrochloric acid
- test tubes
- test-tube rack
- known solids
- unknown solids

Safety

- Wear eye protection.
- Hydrochloric acid and test solutions are irritants.
- Substances on the flame test loop may spit when in the Bunsen burner flame.
- Close the air hole in between tests to give a luminous flame.

Your teacher may watch to see if you can:

- carry out an experiment appropriately.
- use apparatus accurately and safely.

Method

Cleaning the flame test loop

A Light the Bunsen burner and open the air hole to give a hot blue flame.

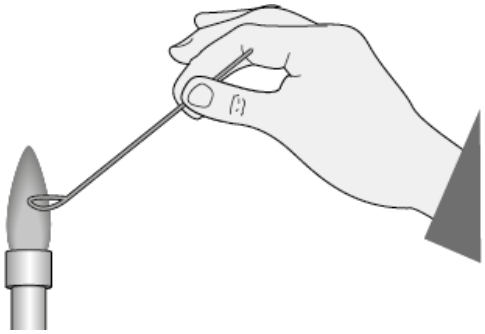
B Dip the flame test loop into the dilute hydrochloric acid. Hold the loop in the flame, then dip it in a beaker of water.

C If you see an intense flame colour in step **B**, repeat step **B** until you only see a faint flame colour or none.

Determining flame test colours

D Dip the clean flame test loop into one of the four known solids (containing Li^+ , Na^+ , K^+ or Ca^{2+} ions).

E Hold the loop in the edge of a hot blue flame, as shown in the diagram. Observe and record the flame test colour produced in the table.



F Test the other known solids, as in steps **D** and **E**, remembering to clean the loop before each test.

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Identifying metal ions

G Carry out flame tests on the four unknown solids. Observe and record each flame test colour.

Recording your results

Record the flame test colours of the known and unknown solids in the tables below.

Metal Ion	Flame test colour
lithium, Li^+	
sodium, Na^+	
potassium, K^+	
calcium, Ca^{2+}	

Unknown	Flame test colour
1	
2	
3	
4	

Considering your results/conclusions

1 Use the flame test colours to identify the metal ions in each unknown solid. Complete the table.

Unknown	Metal ion
1	
2	
3	
4	

Evaluation

2 Explain why you need to clean the flame test loop in between each flame test.

_____

_____

_____

_____

3 Explain whether it was difficult to identify the metal ions in any of the unknown solids.

_____

_____

_____

_____

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Core practical – Identifying ions, Part 2

Pearson Edexcel GCSE (9-1) Chemistry

Name _____ **Class** _____ **Date** _____

Part 2: Precipitate tests

Practical Objective

You will find out the metal hydroxide precipitate colours for aluminium, calcium, copper, iron(II) and iron(III) ions. You will then use these tests to identify the metal ions in different unknown solutions.

Apparatus

- eye protection
- dropping pipettes
- test tubes
- test-tube rack
- dilute sodium hydroxide solution
- known solutions
- unknown solutions

Safety

- Wear eye protection.
- Sodium hydroxide solution is an irritant. Avoid skin contact.

Your teacher may watch to see if you can:

- carry out an experiment appropriately.
- use apparatus accurately and safely.

Method

Determining metal hydroxide precipitate colours

A Using a dropping pipette, fill a test tube to a depth of about 2 cm with one of the five known solutions (containing Al^{3+}, Ca^{2+}, Cu^{2+}, Fe^{2+} or Fe^{3+} ions).

B Using a different dropping pipette, add a few drops of sodium hydroxide solution to the tube. Hold the test tube near the top and shake the bottom gently from side to side to mix its contents.

C Observe and record the colour of the precipitate produced.

D If a white precipitate forms in step **C**, add more sodium hydroxide solution until the test tube is about half full. Observe and record whether the precipitate disappears to leave a colourless solution.

E Repeat steps **A** to **D** with the other known solutions. Do not contaminate one solution with another.

Identifying metal ions

F Carry out steps **A** to **D** on the five unknown solutions. Observe and record the colour of each precipitate and whether any white precipitate formed disappears when excess sodium hydroxide solution is added.

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Recording your results

1 Record the precipitate colours of the known and unknown solutions in the tables below.

Metal ion	Precipitate colour	Effect of adding excess sodium hydroxide
aluminium, Al^{3+}		
calcium, Ca^{2+}		
copper, Cu^{2+}		
iron (II), Fe^{2+}		
iron(III), Fe^{3+}		

Unknown	Precipitate colour
1	
2	
3	
4	
5	

Considering your results/conclusions

2 Use your results to identify the metal ions present in each unknown solution. Complete the table.

Unknown	Metal ion
1	
2	
3	
4	
5	

Evaluation

3 Explain why step **D** is needed to identify some metal ions in solution.

_____

_____

_____

_____

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Core practical – Identifying ions, Part 3

Pearson Edexcel GCSE (9-1) Chemistry

Name _____ **Class** _____ **Date** _____

Part 3: Negative ion tests

Practical Objective

You will use simple chemical tests to identify halide ions (Cl^- , Br^- or I^-), sulfate ions (SO_4^{2-}) or carbonate ions (CO_3^{2-}) in known solutions. You will then use these tests to identify the anions in different unknown solutions.

Apparatus

- eye protection
- dropping pipettes
- test tubes
- test-tube rack
- dilute nitric acid
- silver nitrate solution
- dilute hydrochloric acid
- barium chloride solution
- known solutions
- unknown solutions

Safety

- Wear eye protection and avoid skin contact with the substances used.
- Barium chloride solution is harmful.
- Dilute nitric acid and dilute hydrochloric acid are irritants.

Your teacher may watch to see if you can:

- use apparatus accurately and safely.

Method

Testing for halide ions

A Fill a test tube to a depth of about 2 cm with one of the solutions containing halide ions (Cl^- , Br^- or I^- ions).

B Add a few drops of dilute nitric acid. Hold the tube near the top and shake the bottom gently from side to side to mix its contents.

C Add a few drops of silver nitrate solution. Record the colour of the precipitate produced.

D Repeat steps **A** to **C** with the other known solutions that contain halide ions (Cl^- , Br^- or I^- ions).

Testing for sulfate ions

E Fill a test tube to a depth of about 2 cm with the solution containing sulfate ions, SO_4^{2-}.

F Add a few drops of dilute hydrochloric acid. Hold the tube near the top and shake the bottom gently from side to side to mix its contents.

G Add a few drops of barium chloride solution. Record the colour of the precipitate produced.

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Testing for carbonate ions

H Fill a test tube to a depth of about 2 cm with the solution containing carbonate ions, CO_3^{2-}.

I Add a few drops of dilute hydrochloric acid. Record whether bubbling occurs.

Identifying anions in solution

J Fill a test tube to a depth of about 2 cm with one of the unknown solutions.

K Carry out a test for halide ions or for sulfate ions. Also look for bubbling (which indicates the presence of carbonate ions) when you acidify the solution with dilute nitric or with dilute hydrochloric.

L Repeat steps **J** and **K** until you have determined which anions each unknown solution contains.

Recording your results

Record your results in the tables below:

Testing for halide ions

Halide ion	Silver halide precipitate colour
Cl^-	
Br^-	
I^-	

Unknown	Observations
1	
2	
3	
4	
5	

Testing for sulfate ions

Sulfate ion	Precipitate colour
SO_4^{2-}	

Unknown	Observations
1	
2	
3	
4	
5	

Testing for carbonate ions

Carbonate ion	Bubbling visible (yes/no)
CO_3^{2-}	

Unknown	Observations
1	
2	
3	
4	
5	

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.

Considering your results/conclusions

1 Use your results to identify the anions in each unknown solution. Complete the table below.

Unknown	Anion
1	
2	
3	
4	
5	

Evaluation

2 Silver carbonate and barium carbonate are insoluble solids. Suggest an explanation for why dilute acids are added to the test solutions in steps **B** and **F**.

© Pearson Education Ltd 2024. Copying permitted for purchasing institution only. This material is not copyright free.